GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

Qualification Pack

Electrical Winder

QP Code: PSS/Q4401

Version: 1.0

NSQF Level: 4

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Qualification Pack

PSS/Q4401: Electrical Winder

Brief Job Description

The individual in this job role performs motor winding and re-winding of armature, stator/rotor, transformer including replacement of insulation and varnishing. The individual also disassembles, troubleshoots, performs service and maintenance of electrical power tools.

Personal Attributes

The individual must physically and mentally be able to perform essential functions of the job safely. The job holder must have an eye for detail, patience and discipline to carry out detailed and repetitive tasks. The candidate should be able to read, listen carefully and understand instructions and warnings.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. PSS/N1331: Apply basic health and safety practices for power related work
2. PSS/N1336: Working effectively with others
3. PSS/N4401: Prepare for electrical winding operations
4. PSS/N4402: Perform winding for armature, transformer, AC and DC motors
5. PSS/N4403: Perform basic care and maintenance of hand and power tools for electrical work

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution Downstream
Occupation	Operation and Maintenance - Distribution Downstream
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7412.0400, 8282.50

Minimum Educational Qualification & Experience	8th Class
Minimum Level of Education for Training in School	8th Standard
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	29/05/2019
Next Review Date	29/05/2023
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03489
NQR Version	1.0

PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Scope

This unit/task covers the following:

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

PC1. • use protective clothing/equipment for specific tasks and work conditionsProtective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visorsequipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator

PC2. identify job-site hazards and possible causes of risks/accident at the workplaceHazards- electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions

PC3. follow procedures for safe working with electrical equipmentSafe Working Procedures: tag out/lock out, ptw (permit to work)

PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc

PC5. use standard safe working practices when working at heights, confined areas and trenches

PC6. test any electrical equipment and system using insulated testing devices before touching them

PC7. ensure positive isolation of electrical equipment & system as per the given standards

PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installedParameters: temperature, pressure, flow and current

PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.

PC10. use various methods of accident prevention in the work environmentMethods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors

PC11. identify and retrieve general health and safety equipment in the workplaceGeneral Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.

PC12. set up and use scaffolds, elevated platforms and ladders safelySet up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc

PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify themFaults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.

PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa

PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage

PC16. store flammable materials and machine lubricating oil safely and correctly

PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC18. use good housekeeping practices at all timesGood Housekee Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces

PC19. identify common hazard signs displayed in various areasVarious Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.

PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC21. use various appropriate fire extinguishers for different types of fires correctlyTypes of Fires: class a, b, c, d and e

PC22. use appropriate rescue techniques during fire hazard

PC23. maintain good housekeeping in order to prevent fire hazards

PC24. use a fire extinguisher correctly in case of fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC25. administer appropriate first aid to the victims for the relevant injuriesInjuries: bleeding, burns, choking, electric shock, poisoning etc.

PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments

PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services

PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergenciesEmergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work

PC30. free an affected person from electrocution safely as per the recommended procedure

PC31. move injured people and others to a safe place during emergencies

PC32. write an incident report or dictate a report to another person, and send report to the authorized personIncident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
2. names and location of documents that refer to health and safety in the workplace
3. names and location of people responsible for health and safety in the workplace
4. documents that refer to health and safety in the workplacedocuments: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (e.g. government notices)
5. meaning of hazards and risks
6. health and safety hazards commonly present in the work environment and related precautions
7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
8. possible causes of risk and accident possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness); not taking safety precautions
9. methods of accident prevention
10. safe working practices when working with tools and machines
11. safe working practices while working at various hazardous sites
12. where to find all the general health and safety equipment in the workplace
13. various dangers associated with the use of electrical equipment
14. positive isolation of electrical equipment and system
15. safe handling and disposal of hazardous power plant wastes
16. use of emission and pollution control devices and measures taken to control pollution
17. various safety procedures and equipment used to work at heights, trenches and confined places
18. safe working practices specific to working with electrical equipment and system e.g. lock out/ tag out, ptw, etc.

19. preventative and remedial actions to be taken in the case of exposure to toxic materialstoxic materials: solvents, flux, lead exposure: ingested, contact with skin, inhaledpreventative action: ventilation, masks, protective clothing/ equipment); remedial action: immediate first aid, report to supervisor
20. importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
21. precautionary measures taken to prevent fire accident
22. various causes of firecauses of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.
23. correct technique(s) of using a fire extinguisher
24. different methods of extinguishing fire
25. different materials used for extinguishing firematerials: sand, water, foam, co2, dry powder
26. emergency rescue techniques applied during a fire hazard
27. various types of safety signs and what they mean
28. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
29. rescue techniques applied during fire hazards
30. good housekeeping in order to prevent fire hazards
31. correct use of a fire extinguisher
32. how to free a person from electrocution
33. basic techniques of bandaging
34. perform artificial respiration and the cpr process

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write an accident/ incident report in local language and/or english
2. read and comprehend basic content to read labels, charts, signages
3. read and comprehend basic english to read manuals of operation
4. read an accident/ incident report in local language or english
5. question coworkers appropriately in order to clarify instructions and other issues
6. give clear instructions to coworkers and subordinates
7. remain congenial while discussing and debating issues with co-workers
8. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
9. ask for, provide and receive required assistance where possible to ensure achievement of work-related objectives
10. follow organizational rule-based decision-making process
11. take decisions with systematic course of actions and/or response
12. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
13. build customer relationships and use customer centric approach

14. think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
15. seek appropriate assistance from other sources to resolve problems
16. report problems that cannot be resolved to the appropriate authority
17. identify cause and effect relations in own area of work
18. work systematically and logically to resolve the issues, identify the cause and anticipate unexpected results
19. critically evaluate the organizational processes to improve them

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Follow workplace health and safety practices	**19**	**43**	**-**	**-**
PC1. • use protective clothing/equipment for specific tasks and work conditionsProtective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visorsequipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator	1	2	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplaceHazards-electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions	1	2	-	-
PC3. follow procedures for safe working with electrical equipmentSafe Working Procedures: tag out/lock out, ptw (permit to work)	1	2	-	-
PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc	1	2	-	-
PC5. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC6. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC7. ensure positive isolation of electrical equipment & system as per the given standards	1	1	-	-
PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installedParameters: temperature, pressure, flow and current	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.	1	2	-	-
PC10. use various methods of accident prevention in the work environmentMethods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors	1	2	-	-
PC11. identify and retrieve general health and safety equipment in the workplaceGeneral Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.	1	2	-	-
PC12. set up and use scaffolds, elevated platforms and ladders safelySet up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc	1	3	-	-
PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify themFaults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.	1	3	-	-
PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa	-	3	-	-
PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage	1	3	-	-
PC16. store flammable materials and machine lubricating oil safely and correctly	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	2	-	-
PC18. use good housekeeping practices at all timesGood Housekee Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces	1	2	-	-
PC19. identify common hazard signs displayed in various areasVarious Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.	1	2	-	-
PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	2	-	-
Follow fire safety practices	**3**	**8**	**-**	**-**
PC21. use various appropriate fire extinguishers for different types of fires correctlyTypes of Fires: class a, b, c, d and e	1	2	-	-
PC22. use appropriate rescue techniques during fire hazard	1	2	-	-
PC23. maintain good housekeeping in order to prevent fire hazards	1	2	-	-
PC24. use a fire extinguisher correctly in case of fire	-	2	-	-
Follow emergencies, rescue and first-aid procedures	**8**	**19**	**-**	**-**
PC25. administer appropriate first aid to the victims for the relevant injuriesInjuries: bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments	1	4	-	-
PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergenciesEmergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC30. free an affected person from electrocution safely as per the recommended procedure	1	2	-	-
PC31. move injured people and others to a safe place during emergencies	-	3	-	-
PC32. write an incident report or dictate a report to another person, and send report to the authorized personIncident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified	2	2	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Distribution, Transmission, Generation, Distribution Downstream
Occupation	Operation & Maintenance - Generation Supervision - Distribution Erection, Installation and Commissioning - DistributionSurveying - TransmissionOperation & Maintenance - Transmission
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	11/02/2023
NSQC Clearance Date	22/08/2019

PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette,discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required

PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt

PC3. give information to others clearly, at a pace and in a manner that helps them to understand

PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible

PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks

PC6. display appropriate communication etiquette while working

PC7. display active listening skills while interacting with others at work

PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism

PC9. demonstrate responsible and disciplined behaviors at the workplace

PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. eescalation matrix and procedures for reporting work and employment related issues

KU5. various categories of people that one is required to communicate and co-ordinate with in the organization

KU6. importance of effective communication in the workplace

KU7. importance of teamwork in organizational and individual success

KU8. various components of effective communication

KU9. key elements of active listening

KU10. value and importance of active listening and assertive communication

KU11. barriers to effective communication
KU12. importance of tone and pitch in effective communication
KU13. importance of avoiding casual expletives and unpleasant terms while communicating professional circles
KU14. how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
KU15. importance of ethics for professional success
KU16. importance of discipline for professional success
KU17. what constitutes disciplined behavior for a working professional
KU18. common reasons for interpersonal conflict
KU19. importance of developing effective working relationships for professional success
KU20. expressing and addressing grievances appropriately and effectively
KU21. importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. note the information communicated by the officer incharge
GS2. note down observations (if any) related to the operation/maintenance
GS3. read and interpret the process required for different types of manuals
GS4. read and interpret the flowchart of all parts of an assembly
GS5. read manuals and documents to understand the product-details & how they can be used
GS6. discuss task lists, schedules and activities with the colleague/supervisor
GS7. effectively communicate with the team members
GS8. attentively listen and comprehend the information given by the colleague/supervisor/contractor
GS9. communicate clearly with the colleague on the issues faced during query/fault
GS10. follow colleague/contractor rule-based decision making process
GS11. take decisions with systematic course of actions and/or response
GS12. planning and organization of tasks to meet deadlines
GS13. build customer relationships and use customer centric approach
GS14. seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
GS15. work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
GS16. critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
working with others	**30**	**70**	**-**	**-**
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023

PSS/N4401: Prepare for electrical winding operations

Description

This unit is about the competencies required by an electrical winder to perform preparatory tasks to undertake electrical winding for repair and maintenance of electrical motors.

Scope

This unit/ task covers the following:

- Prepare the work area
- Assess the job requirements

Elements and Performance Criteria

Prepare the work area

To be competent, the user/individual on the job must be able to:

PC1. identify personal protective equipment (ppe) and use the same as per the related working environment

PC2. test single phase or three phase supply by multimeter/test lamp/double test lamp

PC3. organise materials required for the rewinding job and varnishing ensuring ease of access

PC4. ensure that the work area is free from dust, debris, metal shavings and is safe for working

PC5. ensure that the tools and materials required for winding operations are adequately available and fit for use

Assess the job requirements

To be competent, the user/individual on the job must be able to:

PC6. uncover the motor by opening and disassembling the small electrical machine/device safely and place the disassembled parts in a secure and clean area

PC7. test the electrical fan/motor for insulation failure

PC8. open and dismantle the motor to access the stator and armature

PC9. evaluate the damage in the motor to establish if re-winding will be effective in restoring motor operations

PC10. identify the damaged/burnt part of winding in fan/motor

PC11. identify the prerequisites to winding such as former and the coils requirements of different size and shape

PC12. record the winding data like size/gauge of wire, number of turns, coil connection, coil pitch, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organisation
2. organizational norms relevant to own employment and performance conditions

3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and relevant sources for information
5. relevant people and their responsibilities within the work area
6. escalation matrix and procedures for reporting work and employment related issues
7. organizational procedures for requisitioning and indenting materials and equipment
8. organisational procedure for recording and reporting job works and related information
9. importance of working in a clean environment, safe practices and procedures
10. various types of chemicals used in electrical winding operations and hazards of working with them
11. common electricity terminology, such as current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
12. correct use of tools and equipment used in motor winding activities
13. personal productive equipment (ppe) and the health and safety precautions for carrying out motor rewinding work
14. functioning of dc and ac single phase and 3-phase supply and the difference between them with respect to voltage, current and power
15. procedure to test single and three phase supply by multimeter/ test lamp/ double test lamp
16. quantity and application of various types of material required in winding operations
17. risks of having metal shavings in a motor winding environment
18. different types of motors, their parts, structure, functioning and the electrical appliances using various types of motors
19. functioning of ammeters, volt meters, meggers, multi-meters
20. difference between rotary parts and static parts
21. how to handle stripping/winding tools safely
22. bis (bureau of indian standards) rules for winding/rewinding
23. insulating materials as per class of insulation (a/e/b/c/f/h) and reasons for insulation failure in electrical machines
24. terminology used in winding like pole pitch, coil pitch etc.
25. preparation of winding data as per old winding and rating plate of machine
26. procedure followed for preparation of winding table, winding diagram, connection diagram and re-winding of all kind of electric motors like single phase ac motors, pump motors, ceiling fan motors, table fan motors, submersible pump motor, etc.
27. procedure and importance of testing for continuity and insulation
28. basic construction and coil arrangement in primary and secondary side of transformer (single phase and three phase) and the concept of turns and voltage ratio
29. types of wires and strips used for transformer coil and the procedure for coil rewinding - hand and motorized coil winding
30. types of varnish, its application and impregnation methods and its advantages
31. concept and principles of basic arithmetic calculation for calculating dimensional parameters as per the drawing and part programming

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work like winding data, material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/sign boards etc.
5. read safety instructions on machines and devices
6. communicate clearly with team and other staff members
7. discuss task lists, schedules and activities with the team members
8. make appropriate work-related judgments based on laid down policies and procedures
9. plan and organize tasks to meet quantity and quality specifications
10. identify impact of own work and actions on end customer
11. identify changes in conditions, operations, and the environment to anticipate and address problems

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare the work area	**9**	**26**	**-**	**-**
PC1. identify personal protective equipment (ppe) and use the same as per the related working environment	1	4	-	-
PC2. test single phase or three phase supply by multimeter/test lamp/double test lamp	3	7	-	-
PC3. organise materials required for the rewinding job and varnishing ensuring ease of access	3	5	-	-
PC4. ensure that the work area is free from dust, debris, metal shavings and is safe for working	1	5	-	-
PC5. ensure that the tools and materials required for winding operations are adequately available and fit for use	1	5	-	-
Assess the job requirements	**21**	**44**	**-**	**-**
PC6. uncover the motor by opening and disassembling the small electrical machine/device safely and place the disassembled parts in a secure and clean area	3	6	-	-
PC7. test the electrical fan/motor for insulation failure	4	7	-	-
PC8. open and dismantle the motor to access the stator and armature	3	7	-	-
PC9. evaluate the damage in the motor to establish if re-winding will be effective in restoring motor operations	5	8	-	-
PC10. identify the damaged/burnt part of winding in fan/motor	1	4	-	-
PC11. identify the prerequisites to winding such as former and the coils requirements of different size and shape	2	4	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. record the winding data like size/gauge of wire, number of turns, coil connection, coil pitch, etc.	3	8	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N4401
NOS Name	Prepare for electrical winding operations
Sector	Power
Sub-Sector	Distribution Downstream
Occupation	Operation and Maintenance - Distribution Downstream
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	29/05/2019
Next Review Date	29/05/2023
NSQC Clearance Date	22/08/2019

PSS/N4402: Perform winding for armature, transformer, AC and DC motors

Description

This unit is about performing winding operations for motors, transformers, armatures and performing finishing activities on the equipment after winding.

Scope

This unit/ task covers the following:

- Perform winding for motors
- Perform winding of transformers
- Perform rewinding operations for armatures
- Perform finishing activities

Elements and Performance Criteria

Perform winding for motors

To be competent, the user/individual on the job must be able to:

PC1. prepare insulating paper and wooden/insulating stick as per the slot of motor
PC2. prepare the coil as per size, number of turns and coil pitch of the given motor
PC3. insert and dress the insulating material in each slot of the motor
PC4. insert the coil and mark start/ end point
PC5. connect the coils as per connection diagram
PC6. test for continuity and winding insulation
PC7. assemble and run the motor (without varnishing)

Perform winding of transformers

To be competent, the user/individual on the job must be able to:

PC8. measure and determine the size of winding wire/ strip for primary and secondary sides of the transformer
PC9. test and identify the faults in coil of primary and secondary sides of the transformer
PC10. dismantle the core and coil
PC11. repair/rewind the faulty coils
PC12. clean the core and provide suitable insulation
PC13. perform winding using the hand and/or motorized coil winding
PC14. reassemble the coils of primary and secondary sides of the transformer
PC15. make connections and test for continuity and IR (Insulation Resistance) of primary and secondary sides of the transformer
PC16. test the transformer for insulation, transformation ratio and performance

Perform rewinding operations for armatures

To be competent, the user/individual on the job must be able to:

PC17. prepare insulating paper and wooden/insulating stick as per the slot of the armature
PC18. test the armature, coil connections and record the winding data
PC19. strip/remove the armature winding
PC20. clean and insert the insulation in the slots
PC21. prepare and insert the coil in slots
PC22. identify coil ends and make connections to commutator raiser
PC23. properly secure/tie the coil ends on the armature
PC24. test for continuity and ir of winding
PC25. varnish the armature winding

Perform finishing activities

To be competent, the user/individual on the job must be able to:

PC26. varnish the re-winded motor
PC27. examine the insulation of winding using a megger insulation tester
PC28. make proper connections and check the performance of motor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organisation
2. organizational norms relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and relevant sources for information
5. relevant people and their responsibilities within the work area
6. escalation matrix and procedures for reporting work and employment related issues
7. organizational procedures for requisitioning and indenting materials and equipment
8. organisational procedure for recording and reporting job works and related information
9. safe handling of stripping/winding tools and reasons for insulation failure in electrical machines
10. BIS (bureau of indian standards) rules for winding/rewinding
11. functioning of dc (direct current) and ac (alternating current) single phase and 3-phase supply and the difference between them with respect to voltage, current and power
12. types of winding wires and class-wise insulation material (a/e/b/c/f/h)
13. preparation of winding data as per old winding and rating plate of machine
14. procedure followed for re-winding of all kind of electric motors
15. procedures to prepare the winding table, connection diagram, winding diagram for given motor and various methods used for inserting coil into the slots
16. basic construction and coil arrangement in primary and secondary side of transformer (single phase and three phase); turns and voltage ratio
17. types of wires and strips used for transformer coil
18. dos and don'ts for removing core and coil from transformer tank; hand and motorized coil winding; placing insulation between coils and core; and testing the general faults in a transformer coil

19. connection, ir testing and improvement of primary and secondary sides of the transformer
20. types of varnishes, their uses, applications, methods of varnish impregnation and its advantages
21. terminology used and types of armature winding like lap winding, wave winding, pole pitch, coil pitch, back and front pitch, progressive and retrogressive winding etc.
22. procedure for preparing winding data for given armature and dismantling the burnt winding wire
23. procedure for providing insulation, inserting coil and connection to commutator
24. process for impregnation and testing of armature winding and securing coil ends on the armature
25. concept and principles of basic arithmetic calculation, co-ordinate system for calculating dimensional parameters as per drawing and for part programming

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work like winding data, material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/sign boards etc.
5. read safety instructions on machines and devices
6. communicate clearly with team and other staff members
7. discuss task lists, schedules and activities with the team members
8. make appropriate work-related judgments based on laid down policies and procedures
9. plan and organize tasks to meet quantity and quality specifications
10. identify impact of own work and actions on end customer
11. identify changes in conditions, operations, and the environment to anticipate and address problems

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform winding for motors	**7**	**19**	**-**	**-**
PC1. prepare insulating paper and wooden/insulating stick as per the slot of motor	1	3	-	-
PC2. prepare the coil as per size, number of turns and coil pitch of the given motor	1	3	-	-
PC3. insert and dress the insulating material in each slot of the motor	1	2	-	-
PC4. insert the coil and mark start/ end point	1	2	-	-
PC5. connect the coils as per connection diagram	1	3	-	-
PC6. test for continuity and winding insulation	1	3	-	-
PC7. assemble and run the motor (without varnishing)	1	3	-	-
Perform winding of transformers	**9**	**25**	**-**	**-**
PC8. measure and determine the size of winding wire/ strip for primary and secondary sides of the transformer	1	3	-	-
PC9. test and identify the faults in coil of primary and secondary sides of the transformer	1	3	-	-
PC10. dismantle the core and coil	1	2	-	-
PC11. repair/rewind the faulty coils	1	3	-	-
PC12. clean the core and provide suitable insulation	1	2	-	-
PC13. perform winding using the hand and/or motorized coil winding	1	3	-	-
PC14. reassemble the coils of primary and secondary sides of the transformer	1	3	-	-
PC15. make connections and test for continuity and IR (Insulation Resistance) of primary and secondary sides of the transformer	1	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. test the transformer for insulation, transformation ratio and performance	1	3	-	-
Perform rewinding operations for armatures	**10**	**19**	**-**	**-**
PC17. prepare insulating paper and wooden/insulating stick as per the slot of the armature	1	3	-	-
PC18. test the armature, coil connections and record the winding data	2	2	-	-
PC19. strip/remove the armature winding	1	1	-	-
PC20. clean and insert the insulation in the slots	1	2	-	-
PC21. prepare and insert the coil in slots	1	3	-	-
PC22. identify coil ends and make connections to commutator raiser	1	2	-	-
PC23. properly secure/tie the coil ends on the armature	1	1	-	-
PC24. test for continuity and ir of winding	1	3	-	-
PC25. varnish the armature winding	1	2	-	-
Perform finishing activities	**4**	**7**	**-**	**-**
PC26. varnish the re-winded motor	1	2	-	-
PC27. examine the insulation of winding using a megger insulation tester	2	2	-	-
PC28. make proper connections and check the performance of motor	1	3	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N4402
NOS Name	Perform winding for armature, transformer, AC and DC motors
Sector	Power
Sub-Sector	Distribution Downstream
Occupation	Operation and Maintenance - Distribution Downstream
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	29/05/2019
Next Review Date	29/05/2023
NSQC Clearance Date	22/08/2019

PSS/N4403: Perform basic care and maintenance of hand and power tools for electrical work

Description

This unit is about the competencies required for performing basic care and maintenance of hand and power tools used for electrical winding operations.

Scope

This unit/ task covers the following:

- Perform maintenance of tools and equipment

Elements and Performance Criteria

Perform maintenance of tools and equipment

To be competent, the user/individual on the job must be able to:

PC1. dismantle and reassemble electrical power tools used in electrical winding operations safely as per manufacturer's instructions

PC2. identify defects and malfunctions in hand and power tools

PC3. carry out maintenance of hand tools including overhauling and changing of defective parts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures of the organisation
2. organizational norms relevant to employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. job role and responsibilities and relevant sources for information
5. about the people and their responsibilities within the work area
6. importance of taking the work permit before initiating any maintenance and repair activities
7. importance of following good housekeeping and fire prevention procedures
8. importance of following job instructions and defined maintenance procedures
9. importance of reporting problems in a timely manner
10. escalation matrix and procedures for reporting work and employment related issues
11. organizational procedures for requisitioning and indenting materials and equipment
12. common electricity terminology such as current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
13. use and functions of tools and equipment used in electrical winding work and testing, repair and maintenance and if they are suitable, calibrated and operating effectively such as spanners, hammers, chisels, gauge etc.

14. specific health and safety precautions which must be taken for various types of hazards when carrying out repair and maintenance work
15. types of electrical power tools as per their application like hand drilling machine, angle grinder, rotary hammer, sander/polisher, blower, heavy duty cutter, portable cut off saw etc. and how to operate them
16. chemicals, lubricants and oils used in oiling and greasing equipment and components
17. importance of understanding abnormal sounds and vibrations in power tools
18. personal protective equipment (ppe) and clothing that must be worn while working in an electrical workshop such as safety helmet, safety glove, insulated safety shoe, ear plugs, dust suppression masks, etc.
19. troubleshooting technique in electrical power tools - like insulation testing, armature defects, field winding, stator winding defects, noisy operation bearing problem, carbon brush changing, turning the commutator surface
20. principles and methods of preventive and breakdown maintenance

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work like winding data, material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/sign boards etc. read safety instructions on machines and devices
5. communicate clearly with team and other staff members
6. discuss task lists, schedules and activities with the team members
7. make appropriate work-related judgments based on laid down policies and procedures
8. plan and organize tasks to meet quantity and quality specifications
9. identify impact of own work and actions on end customer
10. dentify changes in conditions, operations, and the environment to anticipate and address problems

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform maintenance of tools and equipment	**30**	**70**	**-**	**-**
PC1. dismantle and reassemble electrical power tools used in electrical winding operations safely as per manufacturer's instructions	12	28	-	-
PC2. identify defects and malfunctions in hand and power tools	6	14	-	-
PC3. carry out maintenance of hand tools including overhauling and changing of defective parts	12	28	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N4403
NOS Name	Perform basic care and maintenance of hand and power tools for electrical work
Sector	Power
Sub-Sector	Distribution Downstream
Occupation	Operation and Maintenance - Distribution Downstream
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	29/05/2019
Next Review Date	29/05/2023
NSQC Clearance Date	22/08/2019

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	30	70	-	-	100	15
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N4401.Prepare for electrical winding operations	30	70	-	-	100	25
PSS/N4402.Perform winding for armature, transformer, AC and DC motors	30	70	-	-	100	30
PSS/N4403.Perform basic care and maintenance of hand and power tools for electrical work	30	70	-	-	100	25
Total	**150**	**350**	**-**	**-**	**500**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
CPR	Cardiopulmonary Resuscitation
PPE	Personal Protective Equipment
PTW	Permit To Work
kVA	Kilo Volt Ampere
kW	Kilowatt
kWh	Kilowatt Hour
CO2	Carbon Dioxide

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.